

Executive Summary

This analysis covers every on-chain interaction with the Aave V3 Pool contract on Polygon (0x794a...4814ad) over six months: 18,981 wallets and 158,916 transactions. The headline is not growth — it is a spike and a decay. Active wallets peaked at **12,407 in April**, then fell **-94%** to a partial 716 in September; a seven-day moving average confirms a structural downtrend. What remains is thin but maturing: the **top 1% of wallets (~190) drive ~49% of all transactions**, and new-wallet share fell from 100% (March) to ~21% (September) as a small retained core came to dominate the base.

Two measurement facts shape every decision. First, **value is degenerate at the Pool level** — the Pool contract absorbs ~100% of received native POL (positions move as ERC-20 aTokens), so monetary KPIs read near zero and must be rebuilt from function-level deposits, not Pool-level volume. Second, **cost is trivial** — ~\$0.0073 per transaction (~\$1,158 total) — so the economics problem is measurement, not cost.

Implication: treated as a durable user base, the protocol is small and at risk. Prioritize retaining the committed core and fixing value measurement before any acquisition spend; formalize the concentration and automation risk represented by a handful of power wallets.

Key Findings

- **Activity collapse (High).** MAU fell -94% from the April peak (12,407 → 716 partial Sep); 7-day MA confirms a structural downtrend, not seasonality.
- **Thin, maturing core (High).** 79.5% of senders transacted only once, yet new-wallet share fell 100% → ~21% — a small retained base now dominates.
- **Concentration + degenerate value (Medium).** Top 1% (~190 wallets) ≈ 49% of txs; ~100% of native value sits in the Pool contract (median tx value = 0 POL).
- **Negligible cost (High).** ~\$0.0073/tx (~\$1,158 total); the gas-as-%-of-value ratio and value×gas correlation ($r = -0.05$, $p = 0.75$) are noise.
- **Bot-like timing (Medium).** Activity clusters at 12:00 UTC Thursdays, near-zero weekends — consistent with orchestrated/automated usage.
- **Poor retention (High).** Average month-1 retention 21.1%; the April cohort (58% of the base) retained 4.7%, the March cohort 47%.

Recommendations

#	Recommendation	Priority	Owner	Next step
R-1	Attack the month-1 retention cliff (21.1% → target 30%)	High	Product Manager	30-day onboarding + reactivation campaign over New/Emerging (714) and High-Value Dormant + Frequent (6,257)
R-2	Formalize concentration/dependency risk	High	Operations	Add top-1% share trend to dashboard; alert if >60% in any 7-day window; key-account list from <code>dim_wallets</code>
R-3	Fix value-measurement gap	High	Finance	Decode deposit/borrow/withdraw at function level; load real-time POL-USD + aToken prices; recompute monetary KPIs
R-4	Reframe growth around a durable-base KPI	Medium	Executive	Add “retained core (active ≥2 of last 3 months)” card next to MAU on Page 1
R-5	Keep Polygon; re-measure economics after R-3	Low	Finance	Enforce gas ≤ 0.5% of transferred value by function once R-3 lands
R-6	Build explicit bot/anomaly detection	High	Risk/Compliance	Flag wallets with ≥90% of activity in the peak hour-day cluster; alert on per-wallet rate spikes

Full analysis, methodology, limitations, and evidence backings: [outputs/reports/business_recommendations.md](#).